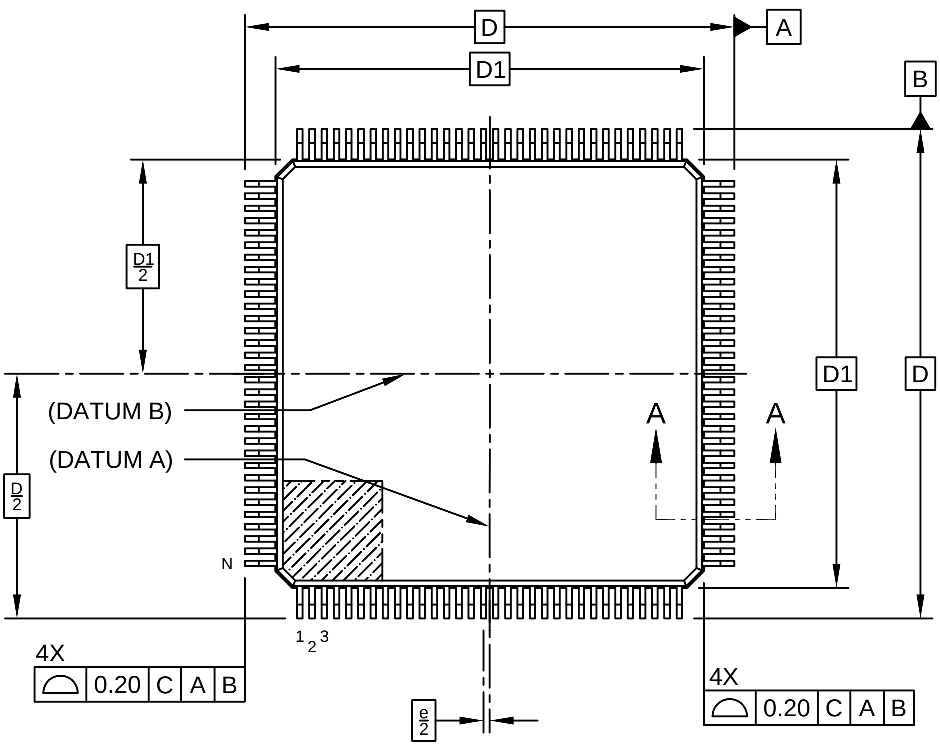
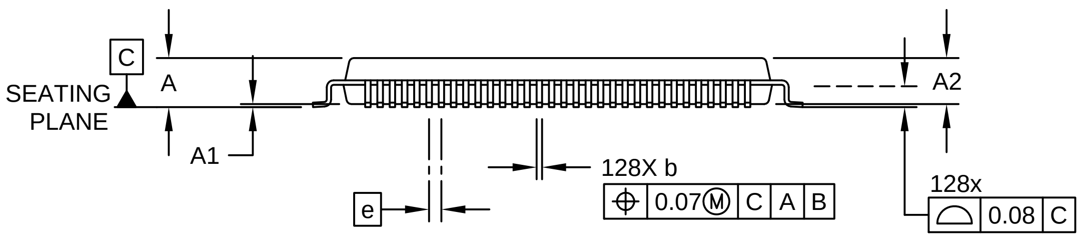
128-Lead Plastic Low Profile Quad Flat Pack (2JB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



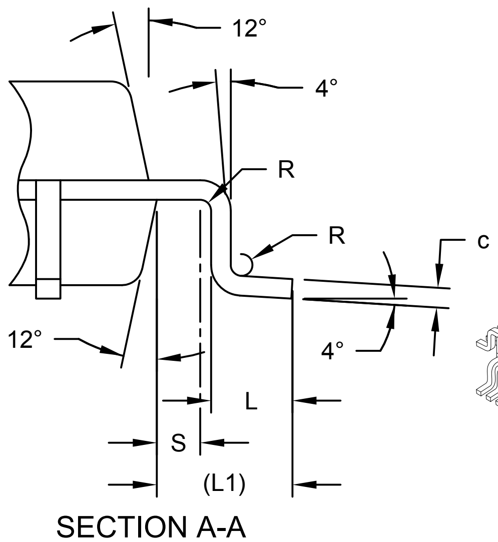
TOP VIEW

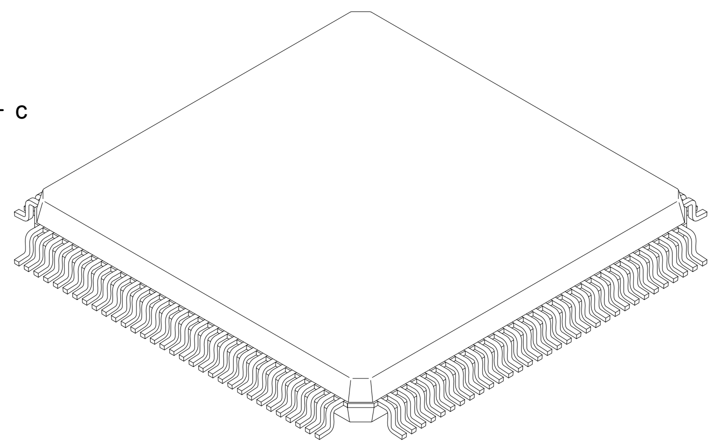


SIDE VIEW

128-Lead Plastic Low Profile Quad Flat Pack (2JB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging





	Units	MILLIMETERS		
	Dimension Limits	MIN	NOM	MAX
Number of Terminals	N		128	
Pitch	e		0.40 BSC	
Overall Height	A	-	-	1.60
Standoff	A1	0.05	-	0.15
Molded Package Thickness	A2	1.35	1.40	1.45
Overall Length	D		16.00 BSC	
Molded Package Length	D1		14.00 BSC	
Overall Width	E		16.00 BSC	
Molded Package Width	E1		14.00 BSC	
Terminal Width	b	0.13	0.18	0.23
Terminal Thickness	c	0.09	-	0.20
Terminal Length	L	0.40	0.60	0.75
Footprint	L1		1.00 REF	-
Lead Bend Radius	R	0.08	-	-
Molded Package to Bend Radius Center	S	0.20	-	-
Foot Angle	ϴ	0°	3.5°	7°
Lead Angle	ϴ1	0°	-	-
Terminal-to-Exposed-Pad	ϴ2	11°	12°	13°

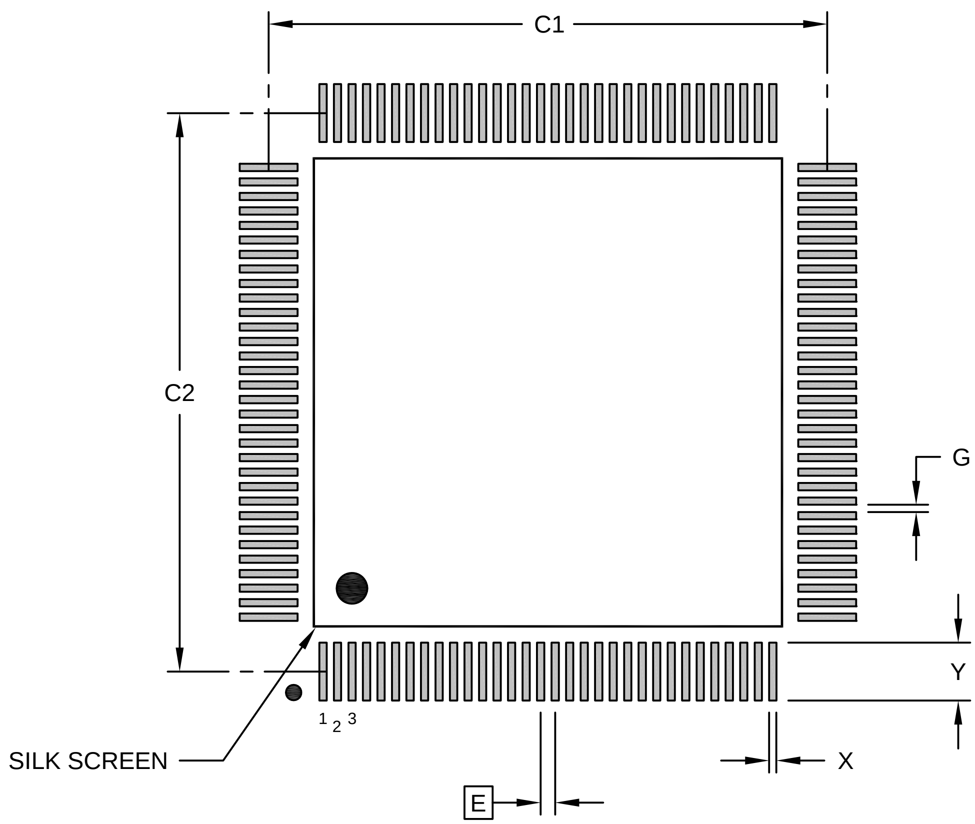
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21006-2JB Rev A Sheet 2 of 2

128-Lead Plastic Low Profile Quad Flat Pack (2JB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Contact Pad Spacing	C1		15.40	
Contact Pad Spacing	C2		15.40	
Contact Pad Width (X128)	X			0.20
Contact Pad Length (X128)	Y			1.50
Contact Pad to Contact Pad (X124)	G	0.20		

Notes:

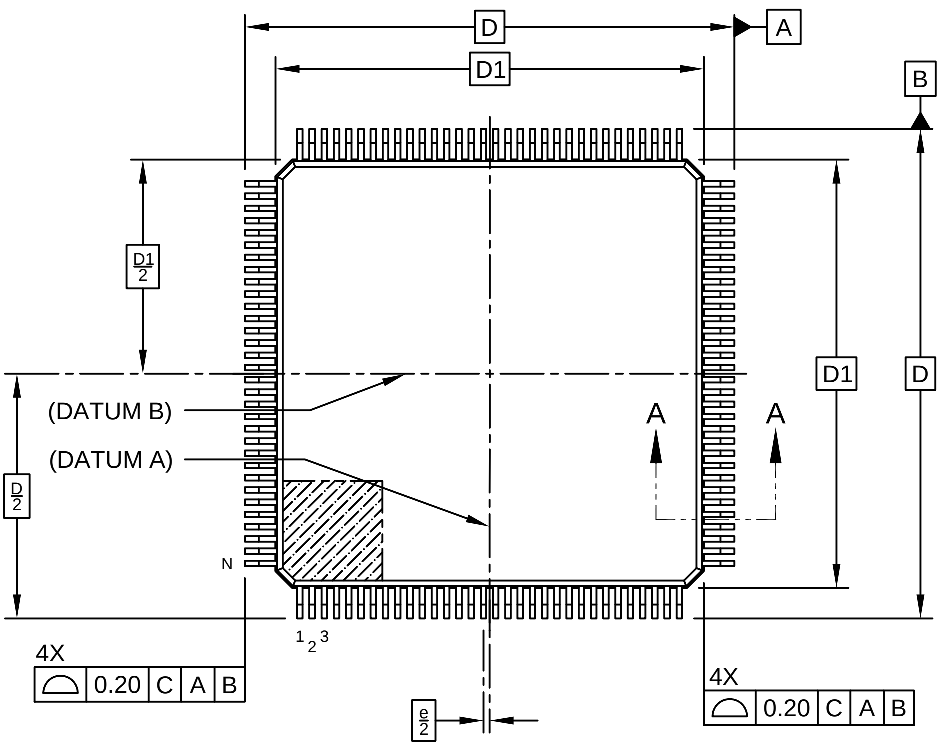
1. Dimensioning and tolerancing per ASME Y14.5M

   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

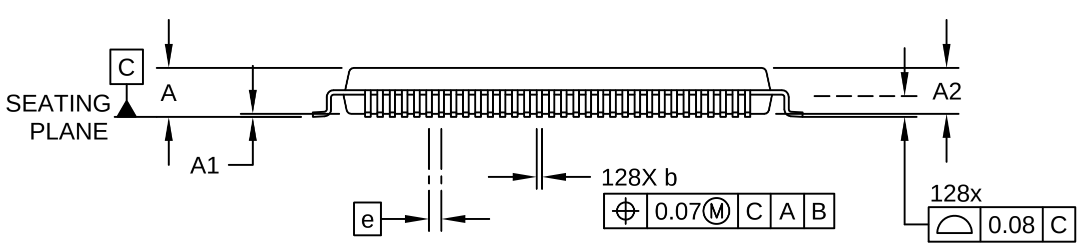
Microchip Technology Drawing C04-23006-2JB Rev A

128-Lead Plastic Low Profile Quad Flat Pack (2KB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



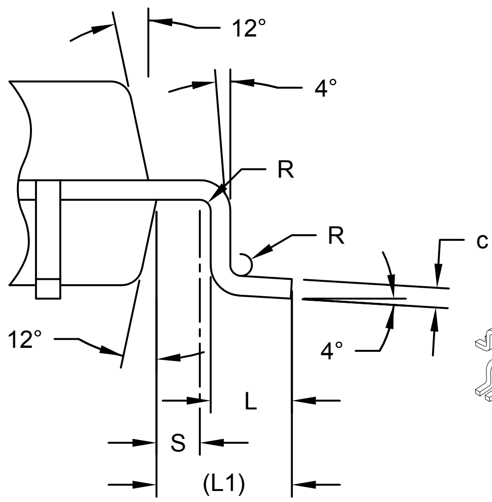
TOP VIEW



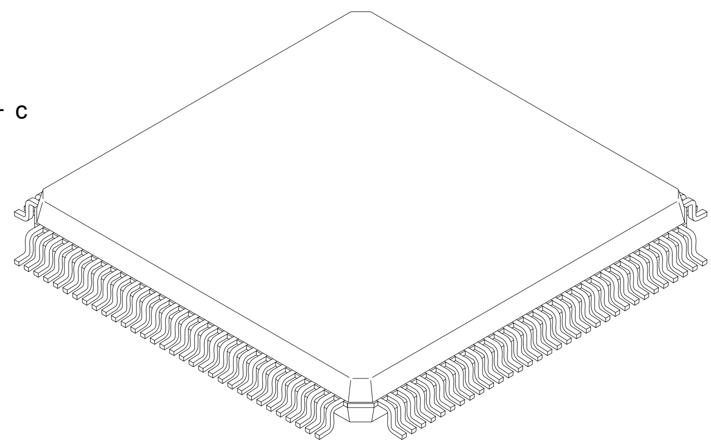
SIDE VIEW

128-Lead Plastic Low Profile Quad Flat Pack (2KB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



SECTION A-A



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		128	
Pitch	e		0.40 BSC	
Overall Height	A	-	-	1.60
Standoff	A1	0.05	-	0.15
Molded Package Thickness	A2	1.35	1.40	1.45
Overall Length	D		16.00 BSC	
Molded Package Length	D1		14.00 BSC	
Overall Width	E		16.00 BSC	
Molded Package Width	E1		14.00 BSC	
Terminal Width	b	0.13	0.18	0.23
Terminal Thickness	c	0.09	-	0.20
Terminal Length	L	0.40	0.60	0.75
Footprint	L1		1.00 REF	-
Lead Bend Radius	R	0.08	-	-
Molded Package to Bend Radius Center	S	0.20	-	-
Foot Angle	Θ	0°	3.5°	7°
Lead Angle	Θ1	0°	-	-
Terminal-to-Exposed-Pad	Θ2	11°	12°	13°

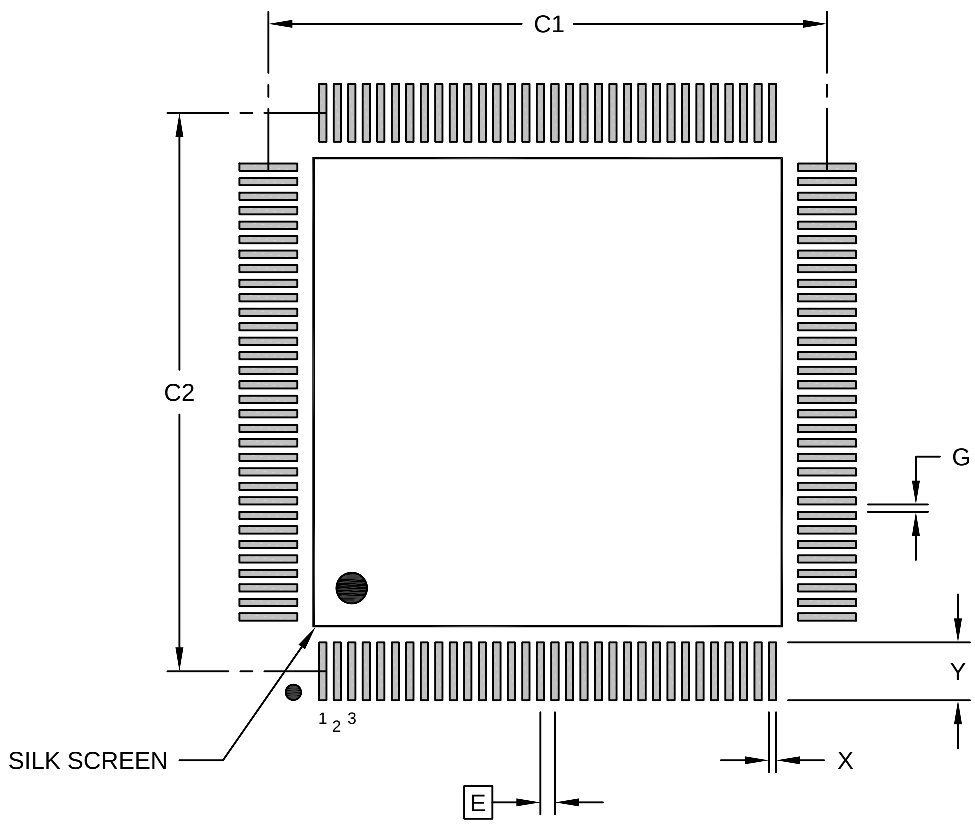
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21006-2KB Rev A Sheet 2 of 2

128-Lead Plastic Low Profile Quad Flat Pack (2KB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note: For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Contact Pad Spacing	C1		15.40	
Contact Pad Spacing	C2		15.40	
Contact Pad Width (X128)	X			0.20
Contact Pad Length (X128)	Y			1.50
Contact Pad to Contact Pad (X124)	G	0.20		

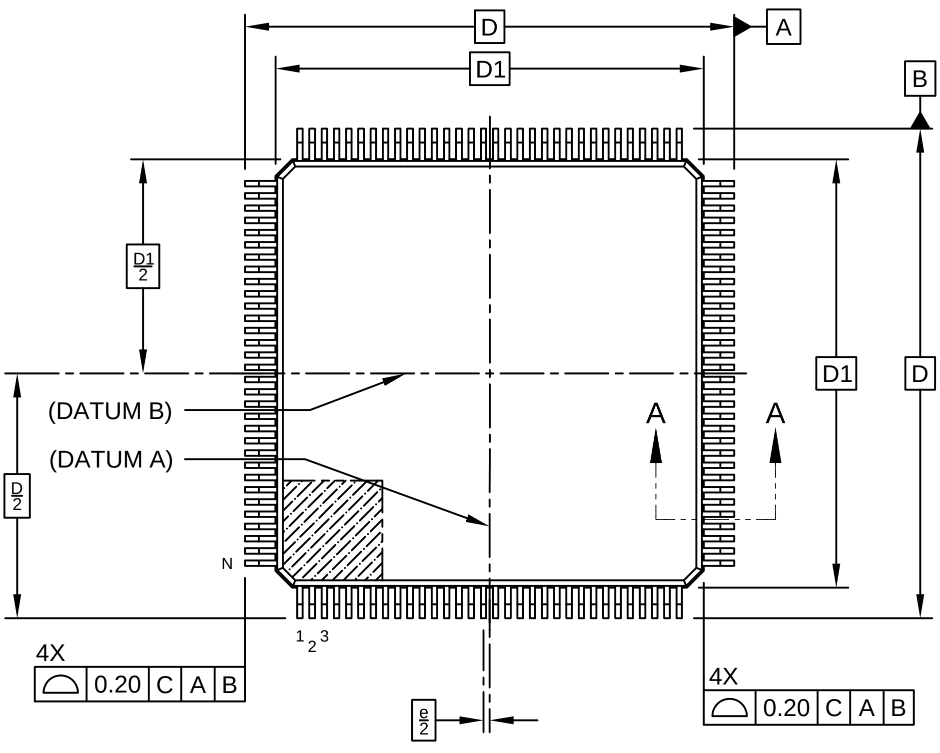
Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

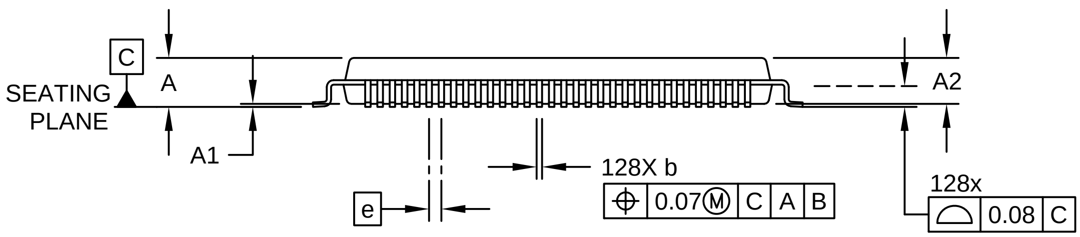
Microchip Technology Drawing C04-23006-2KB Rev A

128-Lead Plastic Low Profile Quad Flat Pack (2LB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



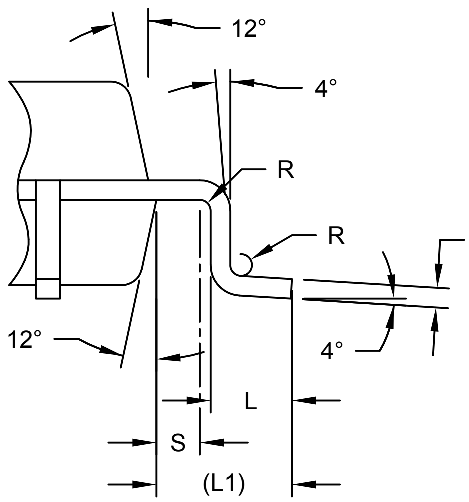
TOP VIEW



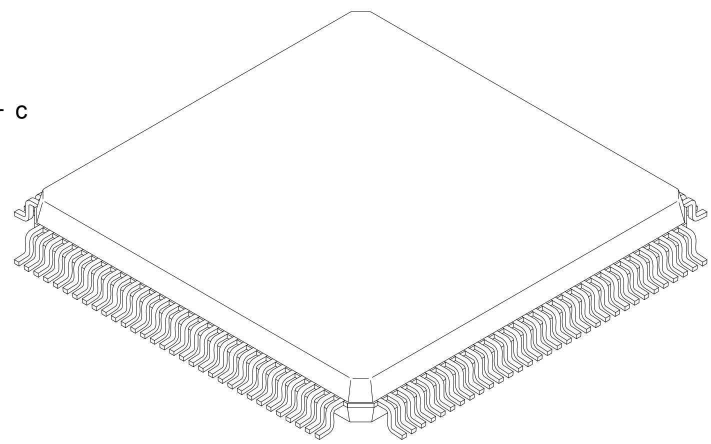
SIDE VIEW

128-Lead Plastic Low Profile Quad Flat Pack (2LB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



SECTION A-A



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		128	
Pitch	e		0.40 BSC	
Overall Height	A	-	-	1.60
Standoff	A1	0.05	-	0.15
Molded Package Thickness	A2	1.35	1.40	1.45
Overall Length	D		16.00 BSC	
Molded Package Length	D1		14.00 BSC	
Overall Width	E		16.00 BSC	
Molded Package Width	E1		14.00 BSC	
Terminal Width	b	0.13	0.18	0.23
Terminal Thickness	c	0.09	-	0.20
Terminal Length	L	0.40	0.60	0.75
Footprint	L1		1.00 REF	-
Lead Bend Radius	R	0.08	-	-
Molded Package to Bend Radius Center	S	0.20	-	-
Foot Angle	Θ	0°	3.5°	7°
Lead Angle	Θ1	0°	-	-
Terminal-to-Exposed-Pad	Θ2	11°	12°	13°

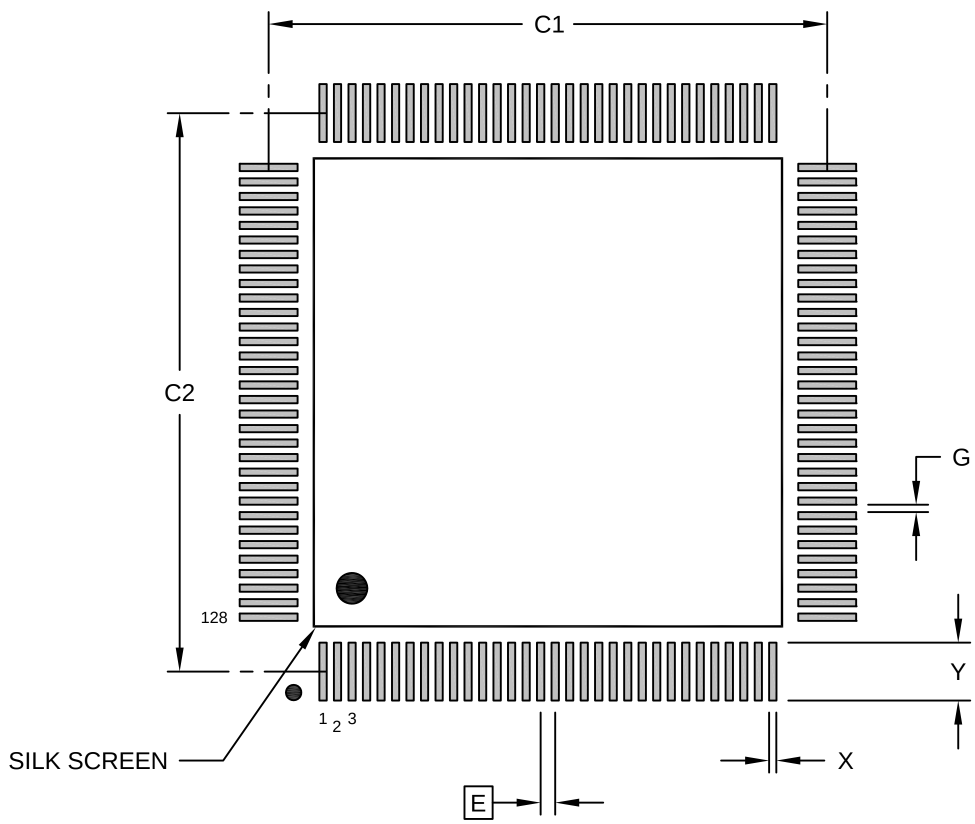
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21006-2LB Rev A Sheet 2 of 2

128-Lead Plastic Low Profile Quad Flat Pack (2LB) - 14x14 mm Body [LQFP]
Atmel Legacy Global Package Code ALN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Contact Pad Spacing	C1		15.40	
Contact Pad Spacing	C2		15.40	
Contact Pad Width (X128)	X			0.20
Contact Pad Length (X128)	Y			1.50
Contact Pad to Contact Pad (X124)	G	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23006-2LB Rev A